

	RV College of Engineering® Department of Computer Science and Engineering CIE - II: Test Paper	
Course & Code	Deep Learning (CS374TFA)	Semester: VII
Scheme & Solutions		

Sl.no.	Part A- Questions	Marks
1.	Teacher forcing is a procedure that emerges from the maximum likelihood criterion, in which during training, the model receives the ground truth output $y^{(t)}$ as input at time $t + 1$. Models that have recurrent connections from their outputs leading back into the model may be trained with teacher forcing.	2
2.	In video activity recognition, a CNN is used to extract spatial features from each video frame, while an RNN models the temporal sequence of these features to recognize actions over time. This combination enables effective learning of both spatial patterns and temporal dependencies, improving classification accuracy.	2
3.	In locally connected layers, each spatial location has its own independent set of weights, resulting in no parameter sharing and high memory requirements . In contrast, tiled convolution uses a limited set of shared filters that repeat across spatial locations, significantly reducing memory usage while still allowing some spatial variation in learned features.	2
4.	The spectrogram is a 2-D data format because it is represented as a matrix with frequency on one axis and time on the other. A convolutional model is suitable because CNNs can effectively learn local spatial patterns in 2-D data, such as time–frequency structures and correlations that characterize different musical notes.	2
5.	Training deep recurrent networks is challenging because gradients must propagate through many time steps and layers, often leading to vanishing or exploding gradients . This results in unstable optimization and difficulty in learning long-term dependencies in sequential data.	2

Sl.no.	Part B- Questions	Marks
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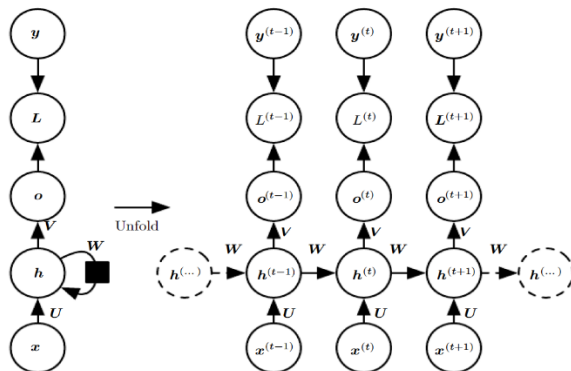
1.	Block diagram of the LSTM recurrent network “cell.” Cells are connected recurrently to each other, replacing the usual hidden units of ordinary recurrent networks. An input feature is computed with a regular artificial neuron unit. Its value can be accumulated into the state if the sigmoidal input gate allows it. The state unit has a linear self-loop whose weight is controlled by the forget gate. The output of the cell can be shut off by the output gate. All the gating units have a sigmoid nonlinearity, while the input unit can have any squashing nonlinearity. The state unit can also be used as an extra input to the gating units. The black square indicates a delay of a single time step. Diagram -5m Explanation with necessary equations -5m 5+5 =10 m	10
2.	1. Standard Convolution 2. Convolution with stride 3. Tiled Convolution 4. Unshared Convolution -2.5 X 4 =10m	10

3.

Important design patterns for recurrent neural networks include

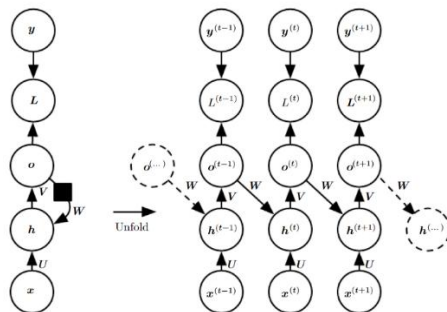
1. Recurrent networks that produce an output at each time step and have recurrent connections between hidden units
2. Recurrent networks that produce an output at each time step and have recurrent connections only from the output at one time step to the hidden units at the next time step
3. Recurrent networks with recurrent connections between hidden units, read an entire sequence and then produce a single output. -1m

Recurrent networks that produce an output at each time step and have recurrent connections between hidden units



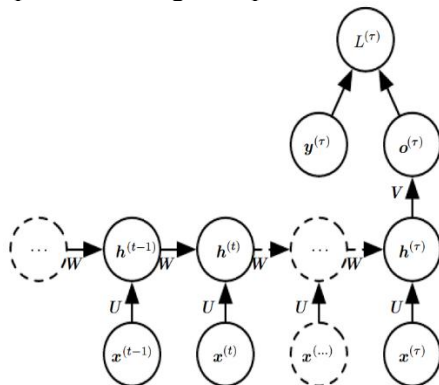
-3m

Recurrent networks that produce an output at each time step and have recurrent connections only from the output at one time step to the hidden units at the next time step.



-3m

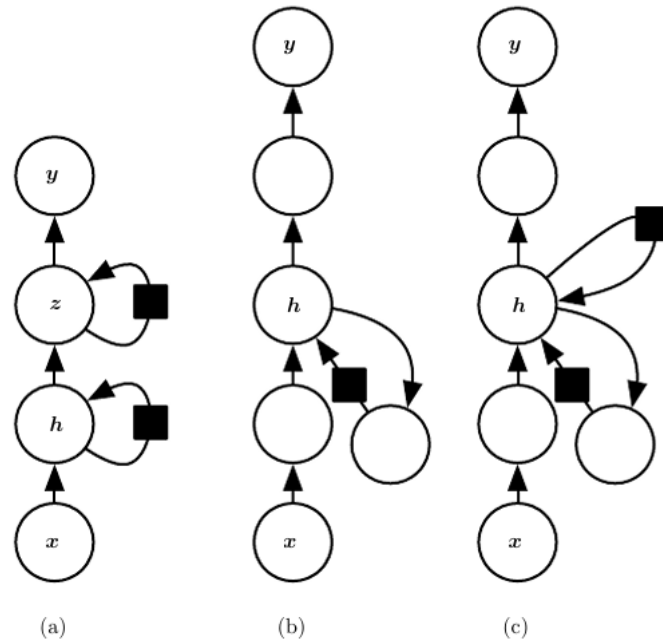
Recurrent networks with recurrent connections between hidden units, read an entire sequence and then produce a single output



-3m

1+3+3+3=10m

4.	RNN can be trained to map an input sequence to an output sequence which is not necessarily of the same length. This comes up in many applications, such as speech recognition, machine translation or question answering, where the input and output sequences in the training set are generally not of the same length (although their lengths might be related). The idea is very simple: (1) an encoder or reader or input RNN processes the input sequence. The encoder emits the context C, usually as a simple function of its final hidden state. (2) a decoder or writer or output RNN is conditioned on that fixed-length vector to generate the output sequence $Y = (y^{(1)}, \dots, y^{(n_y)})$. 5+ 5 =10m	10
5.	The computation in most RNNs can be decomposed into three blocks of parameters and associated transformations: 1. from the input to the hidden state 2. from the previous hidden state to the next hidden state 3. from the hidden state to the output.	10



(a) The hidden recurrent state can be broken down into groups organized hierarchically. (b) Deeper computation (e.g., an MLP) can be introduced in the input-to-hidden, hidden-to-hidden and hidden-to-output parts. This may lengthen the shortest path linking different time steps. (c) The path-lengthening effect can be mitigated by introducing skip connections.

Deep recurrent architectures improve performance over shallow RNNs because:

- **Deeper transformations** extract richer temporal features.
- **Hierarchical hidden layers** learn patterns at multiple time scales, retaining long-term dependencies more effectively.
- **Skip connections** ensure stable training, stronger gradient flow, and preservation of vital information.

Thus, deep RNNs (or variants like deep LSTMs/GRUs) provide significantly better modelling of complex, long-range sequential data.